

5G RRC Signaling Storm Mitigation via a Proactive Digital Twin Architecture with Continuous Calibration

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BLG 632E – Next Generation Wireless Networks

Spring 2025–2026

Outline

- 1 Problem Definition & Research Gap
- 2 System Model
- 3 Proposed Method
- 4 Results & Evaluation
- 5 Conclusion

Motivation: Why 5G Availability Matters

5G NR supports mission-critical services

- Autonomous vehicles require sub-10 ms round-trip latency
- Remote surgery and industrial IoT need **five-nines reliability**
- Every device starts with an RRC connection request through the **gNB RACH**

Consequence of RACH failure

- No RACH access $\Rightarrow$ no data plane
- A single gNB serves hundreds to thousands of UEs simultaneously
- Overloading the RACH blocks **all** connected services

gNB RACH at a Glance

- $c = 54$ preamble channels
- Buffer capacity $K = 200$
- Requests arriving when $Q_{\text{real}} \geq K$ are dropped immediately

Security Exposure

The RACH buffer is a shared, limited resource with no built-in overload anticipation.

What is an RRC Signaling Storm?

RRC and the RACH

RRC is the 3GPP Layer-3 protocol managing connection setup. No data flows without a completed RRC connection.

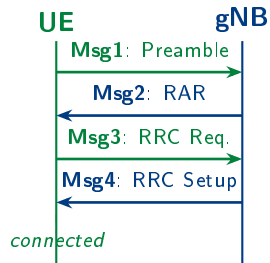
How a storm forms

- Botnet floods Msg1 simultaneously, overwhelming $c = 54$ preamble slots
- Preamble collision: two UEs pick same slot $\Rightarrow$ both fail $\Rightarrow$ retry
- gNB buffers pending Msg3 (RRC Requests) up to $K = 200$
- At $\lambda_{\text{high}} = 500$ req/s, buffer fills in 0.4 s
- Dropped devices retry $\Rightarrow$ load never subsides

Net Effect

RACH becomes a denial-of-service point. Legitimate UEs cannot complete connection setup.

RACH procedure (Msg1–Msg4)

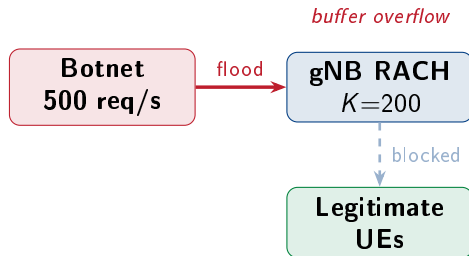


The RRC Signaling Storm Attack

Attack mechanism

- Adversary coordinates a botnet of compromised devices
- Each device sends RRC connection requests at high rate
- Combined rate: $\lambda_{\text{high}} = 500 \text{ req/s}$ vs. normal $\lambda_{\text{low}} = 10 \text{ req/s}$
- RACH buffer saturates within seconds
- Legitimate UE requests are dropped: **QoS failure**

Named *RRC Signaling Storm* in the literature [1].

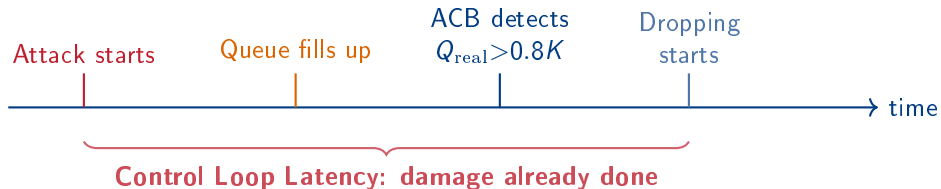


Impact

All connected services lose access until the attack subsides.

Problem: Control Loop Latency in ACB/UAC

Current defence: Access Class Barring (ACB), 3GPP TS 38.331 UAC framework



Fundamental Causality Constraint

ACB throttles **only after** observing $Q_{\text{real}} > 0.8K$. During the build-up, requests are silently dropped.

Empirical Evidence

Kotuliak et al. (2021): RACH exceeded delay budget under high load in commercial 5G NR deployments [6].

Research Gap: Existing Approaches

Reactive ACB / UAC

Strength: Simple, standards-compliant

Limitation: Reacts **after** saturation; inherent control loop latency

ML / DNN Predictors

Strength: High accuracy on stationary traffic

Limitation: $O(W)$ inference cost ($W \sim 10^5 - 10^7$); millisecond latency, impractical on gNB hardware

Standard Digital Twin

Strength: Proactive control possible

Limitation: **Model drift** under non-stationary traffic; state mirroring error grows unbounded

Research Gap

No existing DT approach suppresses model drift with an **online calibration loop** during the attack interval. This work closes that gap.

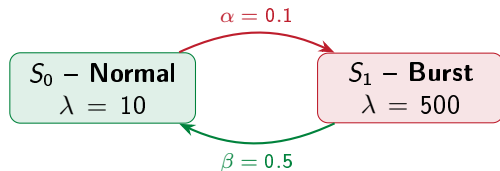
System Model: MMPP/M/c/K Queue

Two-state CTMC (Markov Modulated Poisson Process)

Generator matrix:

$$Q = \begin{bmatrix} -\alpha_{\text{mmp}} & \alpha_{\text{mmp}} \\ \beta_{\text{mmp}} & -\beta_{\text{mmp}} \end{bmatrix}$$

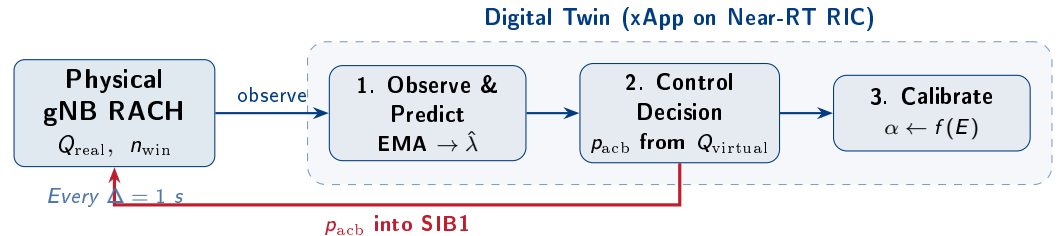
- S_0 (normal): $\lambda_{\text{low}} = 10$ req/s
- S_1 (burst): $\lambda_{\text{high}} = 500$ req/s
- $\alpha_{\text{mmp}} = 0.1 \text{ s}^{-1}$, $\beta_{\text{mmp}} = 0.5 \text{ s}^{-1}$
- Sojourn time $\sim \text{Exp}(|Q_{ii}|)$ – exact CTMC dynamics in DES, no matrix exponential needed



Physical gNB RACH

Preamble channels	$c = 54$
Buffer capacity	$K = 200$
Service time	$\text{Exp}(1)$

Proposed Method: Calibrated Digital Twin



Step 1
Count arrivals in window
 $\Rightarrow$ EMA predicts $\hat{\lambda}$

Step 2
Compute minimum barring
probability
to keep $Q_{\text{real}} < K$

Step 3
Adjust α via asymmetric
step rule if $E > \varepsilon$

EMA Predictor and ACB Control Policy

EMA arrival rate predictor

$$\text{EMA}_t = \alpha \cdot \lambda_{\text{cur}} + (1 - \alpha) \cdot \text{EMA}_{t-1}$$

$$\lambda_{\text{cur}} = n_{\text{win}} / \Delta$$

- Initial weight $\alpha_0 = 0.2$ (validated for bursty traffic [8])
- α is **dynamic**, updated by the calibration loop

Stochastic barring probability

$$p_{\text{acb}} = \max\left(0, \min\left(1, 1 - \frac{K - Q_{\text{real}}}{\max(\text{EMA}_t \cdot \Delta, 10^{-9})}\right)\right)$$

- Minimum probability to prevent $Q_{\text{real}} \geq K$ next step
- Applied as Bernoulli trial per request
- Maps to uac-BarringInfo in SIB1, 3GPP TS 38.331 UAC

Key Insight

p_{acb} rises **before** Q_{real} reaches K : proactive throttling, not reactive.

Continuous Calibration Loop

State Mirroring Error

$$E(t) = |Q_{\text{real}}(t) - Q_{\text{virtual}}(t)|$$

Asymmetric step rule

$$\alpha \leftarrow \begin{cases} \min(\alpha_{\max}, \alpha + \delta^+) & E > \varepsilon \\ \max(\alpha_{\min}, \alpha - \delta^-) & \text{otherwise} \end{cases}$$

Parameters: $\varepsilon = 10$, $\delta^+ = 0.05$, $\delta^- = 0.01$

Why asymmetric?

Attack onset: $\delta^+ = 0.05$

Fast increase: EMA becomes high-frequency.
Outdated observations discarded quickly.

Normal traffic: $\delta^- = 0.01$

Slow decay: statistical stability preserved.
Avoids oscillation around ε .

$\delta^+ \gg \delta^-$: rate mismatch is intentional.

Algorithm 1: Calibrated DT Control Loop

Input: $Q_{\text{real}}, n_{\text{win}}, \Delta, \varepsilon, \alpha, \alpha_{\min}, \alpha_{\max}, \delta^+, \delta^-$

Output: p_{acb} , updated Q_{virtual} , updated α

Every Δ step:

- 1: $\lambda_{\text{cur}} \leftarrow n_{\text{win}} / \Delta$
- 2: $\text{EMA}_t \leftarrow \alpha \cdot \lambda_{\text{cur}} + (1 - \alpha) \cdot \text{EMA}_{t-1}$
- 3: $d \leftarrow \max(\text{EMA}_t \cdot \Delta, 10^{-9})$
- 4: $p_{\text{acb}} \leftarrow \max(0, \min(1, 1 - (K - Q_{\text{real}}) / d))$
- 5: $Q_{\text{virtual}} \leftarrow \text{clip}(Q_{\text{virtual}} + \text{EMA}_t(1 - p_{\text{acb}})\Delta - c\Delta, 0, K)$
- 6: $E \leftarrow |Q_{\text{real}} - Q_{\text{virtual}}|$
- 7: **if** $E > \varepsilon$: $\alpha \leftarrow \min(\alpha_{\max}, \alpha + \delta^+)$
- 8: **else**: $\alpha \leftarrow \max(\alpha_{\min}, \alpha - \delta^-)$
- 9: $\text{drop_rate} \leftarrow p_{\text{acb}}$

$O(1)$ per control step

- Fixed arithmetic operations
- 4 state variables:
 $\{\text{EMA}_t, Q_{\text{virtual}}, \alpha, n_{\text{win}}\}$
- **$O(1)$ time, $O(1)$ space**

vs. DNN predictors

$O(W)$, $W \sim 10^5 - 10^7$

Latency: milliseconds

gNB requires: **microseconds**

Simulation Setup

Experimental configuration

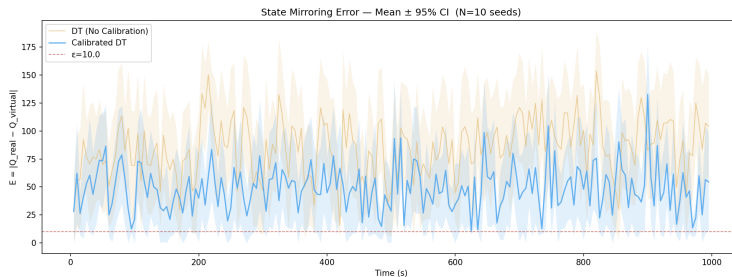
- Discrete-event simulation: SimPy, Python 3.10
- $T_{\text{sim}} = 1000$ s per run
- $N = 10$ independent seeds (Monte Carlo)
- 95% CI: $\bar{x} \pm 1.96 \hat{\sigma} / \sqrt{N}$

Key parameters

Parameter	Symbol	Value
Preamble channels	c	54
Buffer capacity	K	200
Normal arrival rate	λ_{low}	10 req/s
Burst arrival rate	λ_{high}	500 req/s
Control interval	Δ	1 s
Calibration threshold	ε	10
Initial EMA weight	α_0	0.20

Three methods compared: **Reactive Baseline** (threshold $0.8K$, drop 0.5) **DT no-calib.** (α fixed) **Calibrated DT** (asymmetric step rule)

Result 1: State Mirroring Error – 42% Reduction



E_{mean} ($N = 10$, 95% CI)

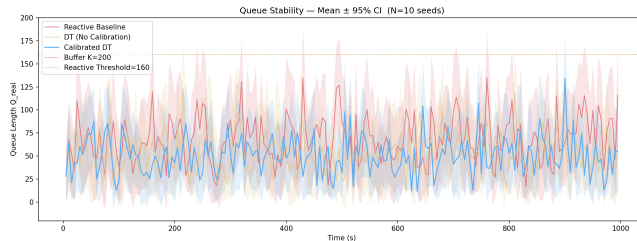
Method	E_{mean}
DT (no calib.)	85.03 ± 4.40
Calib. DT	49.14 ± 3.09

Key Result

$(85.03 - 49.14)/85.03 \approx 42\%$ reduction.

Calibration suppresses model drift continuously during burst.

Result 2: Queue Stability



$P_{success}$ (N = 10, 95% CI)

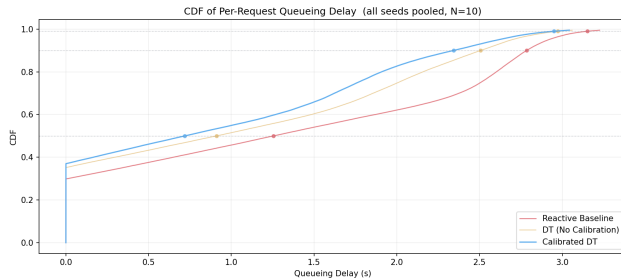
Method	$P_{success}$
Reactive	0.310 ± 0.015
DT (no calib.)	0.301 ± 0.011
Calib. DT	0.309 ± 0.019

Interpretation

Reactive reacts only after $Q_{real} > 0.8K$.

Calibrated DT raises p_{acb} **before** saturation: queue stays in stable region.

Result 3: Queueing Delay CDF



CDF Left-Shift

Calibrated DT curve is shifted left: lower delay at every percentile.

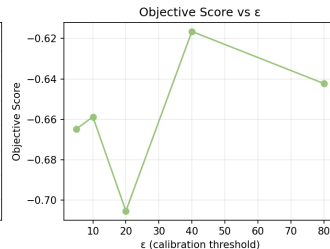
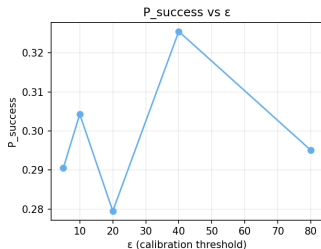
Why tail is suppressed

Reactive allows queuing near saturation: large P90/P99 spikes.

Calibrated DT holds Q_{real} in stable region: more requests served at lower delay.

Result 4: ϵ Sensitivity Analysis

Epsilon Sensitivity Analysis (Calibrated DT, seed=42)



Tested: $\epsilon \in \{5, 10, 20, 40, 80\}$

$\epsilon = 10$ achieves best normalised objective score across all seeds.

Interpretation

Too small ($\epsilon = 5$): over-calibrates in normal traffic.

Too large ($\epsilon = 80$): calibration rarely triggers.

$\epsilon = 10$ is the empirically optimal balance.

O(1) Complexity and O-RAN Deployment

Complexity comparison

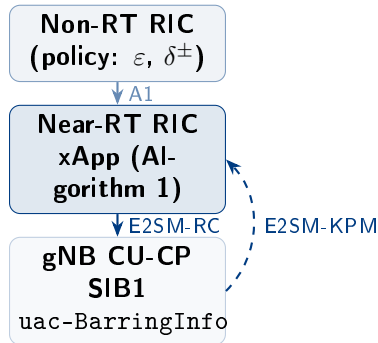
Calibrated DT: O(1) per step

- Fixed arithmetic, 4 state variables
- Completes in nanoseconds on gNB hardware
- gNB RACH deadline: microseconds; ample margin

DNN predictors: O(W)

$W \sim 10^5 - 10^7$ weights. Inference latency: milliseconds. **Impractical on gNB hardware.**

O-RAN integration



- $\Delta = 1$ s fits Near-RT RIC range (10 ms to 1 s)
- No protocol modifications required
- Vendor-independent xApp

Conclusion and Future Work

Three key results

- ❶ **42% reduction** in state mirroring error (calibrated vs. uncalibrated DT)
- ❷ $P_{success}$ **statistically equivalent** to reactive baseline; P90/P99 queueing delay **reduced**
- ❸ **O(1) complexity**: deployable within commercial gNB software via O-RAN xApp

Central Finding

Proactive control in non-stationary 5G environments requires both a sound control policy **and** tight virtual-physical alignment.

Future work

- **Multi-cell extension**: current model treats one gNB cell independently. Real deployments involve handover correlations and inter-cell interference. Requires a distributed DT architecture with inter-cell state sharing.
- **RL-based calibration**: fixed δ^+ , δ^- are vulnerable to slow ramp-up attacks. An RL agent can learn these step sizes dynamically using state mirroring error as a reward signal.

Questions?

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References I

-  D. Rupprecht et al., “Breaking LTE on Layer Two,” *Proc. IEEE S&P*, May 2019, pp. 1121–1136.
-  B. H. Lee et al., “Enhanced Random Access for Massive-MTC,” *IEEE Internet of Things J.*, vol. 8, no. 8, pp. 7046–7064, Apr. 2021.
-  M. Grieves and J. Vickers, “Digital Twin: Mitigating Unpredictable Emergent Behavior,” Springer, 2017.
-  3GPP, “NR; RRC Protocol Specification,” TS 38.331, Rel. 17.3.0, 2022.
-  L. Miuccio et al., “Joint Control of Random Access for Massive MTC in 5G NR,” *IEEE Internet of Things J.*, vol. 7, no. 6, Jun. 2020.
-  M. Kotuliak et al., “Performance Evaluation of 5G NR Non-Standalone Network,” *IEEE Access*, vol. 9, Aug. 2021.
-  S. K. Sharma and X. Wang, “Toward Massive MTC in Ultra-Dense Cellular IoT Networks,” *IEEE Commun. Surveys Tuts.*, vol. 22, no. 1, 2020.

References II

-  R. Dong et al., “Deep Learning for Radio Resource Allocation in 5G,” *IEEE Trans. Wireless Commun.*, vol. 20, no. 4, Apr. 2021.
-  C. Zhang et al., “Deep Learning in Mobile and Wireless Networking: A Survey,” *IEEE Commun. Surveys Tuts.*, vol. 21, no. 3, 2019.
-  H. X. Nguyen et al., “Digital Twin for 5G and Beyond,” *IEEE Commun. Mag.*, vol. 59, no. 2, Feb. 2021.
-  M. Polese et al., “Understanding O-RAN,” *IEEE Commun. Surveys Tuts.*, vol. 25, no. 2, 2023.